

Interference

The redistribution / non-uniform distribution of the light intensity due to superposition of ~~these~~ waves that are in same phase or having same constant phase difference.

on basis of Intensity distribution there are two type of Interference

- Constructive Interference (Maximum Intensity)
- Destructive Interference (Minimum Intensity / zero intensity)

on basis of wave behaviour like

$$y = A \sin \omega t$$

↓ ↘
amplitude phase

- Interference by the division of wavefront
- Interference by or due to division of amplitude
- Examples: - YDS → Young's Double Slit Exp., Fresnel's biprism.
- Examples: - Newton's Ring, Michelson's Interferometer.

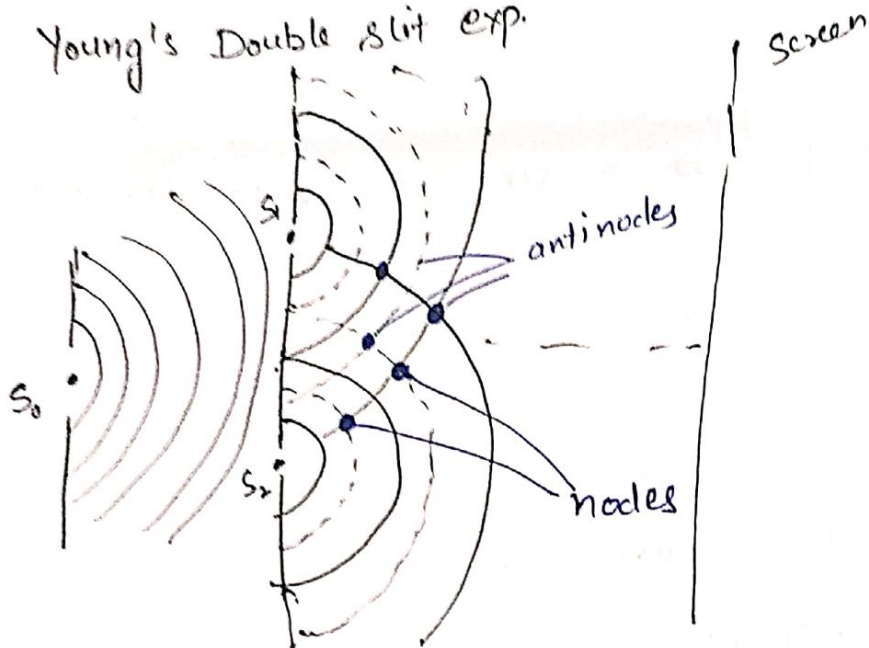
* wavefront :→ A wavefront is the continuous locus of all such particles of the medium which are vibrating in the same phase at any instant.

shapes of wavefront → Shapes of wavefront depend on the source of disturbance.

- like
1. Spherical wavefront → pt. source
 2. Cylindrical " → slit source
 3. Plane wavefront

Huygen's Principle of Secondary wavelets :→
Every pt. of primary wavelet is the source of secondary waves.

Young's Double Slit exp.



→ Coherence: →

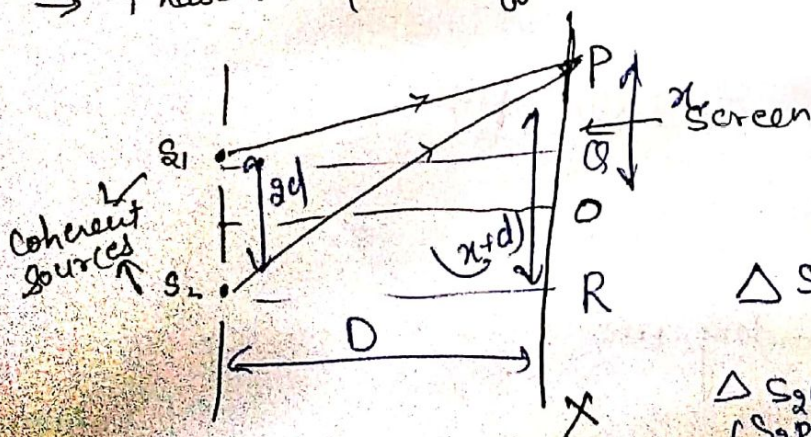
- Temporal coherence
- Spatial coherence

If the phase difference at a single point in the group of waves propagating in space at the beginning and end of fixed time interval does not change with time, then it is temporal coherence.

→ The continuity and uniformity of a light wave in a direction perpendicular to the directions of propagation is spatial coherence.

$$\Delta t = \Delta x / c \rightarrow \text{coherence length}$$

→ Phase & Path difference.



$$\text{Phase} = \frac{2\pi}{\lambda} \times \text{Path diff.}$$

$$\Delta S_1 P = (S_1 P)^2 = (S_1 O)^2 + (OP)^2$$

$$\Delta S_2 P = (S_2 P)^2 = (S_2 R)^2 + (RP)^2 = D^2 + (x-d)^2$$

$$\Rightarrow 2D (S_2 P - S_1 P) \approx 4xd = \frac{2\lambda d}{D}$$

Now conditions for constructive and destructive Interference in YDS Exp. :- $(2d)x = n\lambda$ - max $(n=0) \rightarrow$ bright

let y_1 and y_2 are two wave from S_1 and S_2
 but $S_2P > S_1P$
 so $\underbrace{S_2P - S_1P}_{\text{Path. diff.}} > 0 \Rightarrow \phi = \frac{2\pi}{\lambda} \times (S_2P - S_1P)$
 $\beta = \frac{\lambda D}{2d}$

So $\rightarrow y_1 = a_1 \sin \omega t$

$y_2 = a_2 \sin(\omega t + \phi)$

$\rightarrow$ Resultant $y = y_1 + y_2 = a_1 \sin \omega t + a_2 \sin(\omega t + \phi)$
 $= a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi$
 $= \sin \omega t (a_1 + a_2 \cos \phi) + a_2 \cos \omega t \sin \phi$
 $\downarrow \quad \downarrow$
 $R \cos \theta \quad R \sin \theta$
 $= R \sin \omega t \cos \theta + R \cos \omega t \sin \theta$

$\boxed{y = R \sin(\omega t + \theta)}$ $\rightarrow$ final wave
 $\rightarrow$ Resultant wave with amplitude R

Now amplitude R

$R^2 \cos^2 \theta + R^2 \sin^2 \theta = (a_1 + a_2 \cos \phi)^2 + (a_2 \sin \phi)^2$

$\boxed{R^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi}$

$\downarrow$
 Intensity $I = R^2$

$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$
 Intensity of sustained $a_1, a_2 \propto a$

Then $I = 2a^2 (1 + \cos \phi) = 2a^2 \cdot 2 \cos^2 \phi/2$
 $= 4a^2 \cos^2 \phi/2$

Now for Constructive Interference:-

$$I = 4a^2 \cos^2 \phi/2$$

$$\cos \phi = +1$$

when phase diff $\phi = 2n\pi$, $n = 0, 1, 2, 3, \dots$

$$\text{Path diff.} = \frac{\lambda}{2\pi} \times 2n\pi = n\lambda$$

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi = I_1 + I_2 + 2\sqrt{I_1 I_2}$$
$$= a_1^2 + a_2^2 + 2a_1 a_2$$

$$I_{\max} > I_1 + I_2$$

$$I_{\max} = (a_1 + a_2)^2 = 4a^2$$

Destructive Interference:-

$$\cos \phi = -1 \quad \text{then} \quad \phi = (2n+1)\pi, \quad n = 0, 1, 2, \dots$$

$$\frac{2\pi}{\lambda} \times \text{Path diff} = (2n+1)\pi$$

$$\frac{2\pi}{\lambda} \times (2n+1) \frac{\lambda}{2} = (2n+1)\pi$$

$$I_{\min} = a_1^2 + a_2^2 - 2a_1 a_2 = (a_1 - a_2)^2$$

$$= I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$I_{\min} < (I_1 + I_2)$$

$$\text{If } a_1 \neq a_2$$

$$I_{\min} > 0$$

Visibility of fringes :-

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

$$\frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

$$I_{av} = \frac{\int_0^{2\pi} I \cdot d\phi}{\int_0^{2\pi} d\phi} = \frac{\int_0^{2\pi} (a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi) d\phi}{\int_0^{2\pi} d\phi}$$

$$= \frac{[a_1^2 \phi + a_2^2 \phi + 2a_1 a_2 \sin \phi]_0^{2\pi}}{2\pi} = \frac{2\pi(a_1^2 + a_2^2)}{2\pi} = a_1^2 + a_2^2$$

Shape of Interference fringes

$$S_2 P - S_1 P = n\lambda \quad (\text{maxi})$$

$$(2n+1)\frac{\lambda}{2} \quad (\text{mini})$$

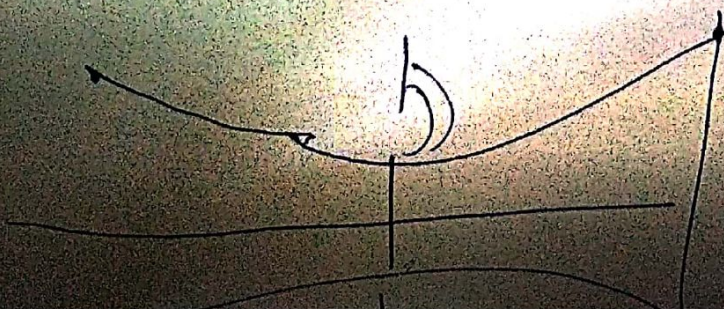
$$\lambda = 10^{-7} \text{ m}$$

$$S_2 P - S_1 P = \text{Const.}$$

this is equation of hyperbola with S_1 & S_2 foci of hyperbola.

$S_2 P - S_1 P$ is also character.

therefore, the eccentricity of fringes is quite large and hence these hyperbola appear, more or less as straight lines.



Fringes with white light are coloured, light is polychromatic and ranging from 4000 Å to 7000 Å and centre is always white.